Recall: If

$$\frac{dy}{dx} = \frac{f(x)}{g(y)},$$

then

$$\int g(y)\, dy = \int f(x)\, dx.$$

Equilibria

Question Consider the differential equation $\frac{dy}{dt} = \frac{1}{5}(4 - y)$.

1. Solve this differential equation.

2. Check that the constant function $y = 4$ is a solution. What value of $y(0)$ does it correspond to?

3. A constant function $y = c$ has $\frac{dy}{dt} =$ ___. Keeping this in mind, describe how to find constant solutions to differential equations from the equation itself.

   .

4. Draw a quick sketch of the slopefield for the differential equation above, and draw two solutions with different behaviors.

5. Describe how to find constant solutions from a slope field for a differential equation.

Definition A constant solution to a differential equation is called an *equilibrium solution.*

Question Find all equilibrium solutions of the following differential equations:

1. $\frac{dz}{dw} = wz(w-1)(z-1)(z+1)$.

2. $\frac{du}{dr} = r\tan(u)$.

Stability

Questions

1. Copy the slopefield for $\frac{dy}{dt} = \frac{1}{5}(4-y)$ from the previous page with its various solutions. Next to it, sketch the slopefield for $\frac{dy}{dt} = \frac{1}{5}(y-4)$, and add in solutions with two different behaviors.

2. Both the above differential equations have an equilibrium at $y =$ ___. However, the solutions nearby behave differently. How so?

Definition An equilibrium solution is *stable* if **all** nearby solutions tend toward it. Otherwise, the equilibrium is *unstable.*

Question If $\frac{dy}{dt} = \frac{1}{5}(4-y)$, the equilibrium at $y =$ ___ is _____. If $\frac{dy}{dt} = \frac{1}{5}(y-4)$, the equilibrium at $y =$ ___ is _____.

Note: The easiest way to determine stability of an equilibrium is to make a rough sketch of the slopefield.

Exercises

For each of the following differential equations, find all equilibrium solutions (if any exist) and determine their stability. Then solve the differential equation (or initial value problem) by separation of variables.

1. $\frac{dy}{dx} = \frac{x}{y}, y(2) = 4$

2. $\frac{dy}{dx} = y^2 x$

3. $\frac{dy}{dx} = -(y-1)^3\sqrt{x}$

4. $\frac{dy}{dx} = x(1 + y^2)$

5. $\frac{dy}{dx} = \sqrt{y}x^2, y(0) = 4$ (Bonus: Could $y(0) = -4$? Explain.)

6. $\frac{dy}{dx} = \cot(y)x$ (Hint: What is another way to write $\cot(y)$?)

You can't always solve the equation!

1. Consider the differetial equation $\frac{dy}{dt} = (y+1)(y-1)(y+2)$.

   (a) Show that this equation is seperable, but that actually solving it is not practical.

   (b) What are the equilbria of this equation?

   (c) By sketching a slopefield, determine the stability of each of the equilibria.

   (d) On your slopefield above, sketch three non-constant solutions with different behaviors.

2. Do the same for the differential equation $\frac{dy}{dt} = ty(y-2)^2(y+2)$. Be careful to account for the fact that t is part of the derivative when sketching your slopefield!